

Alex Biuckians

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SUMMARY

M.S. Data Science student who builds machine-learning systems end to end, from rigorous, leakage-safe modeling through deployed, documented APIs, and defends every result honestly against a baseline. Experienced across supervised, causal, time-series, and anomaly-detection problems, with a track record of shipping containerized services to live URLs, wiring up CI, and reporting skill scores and confidence intervals rather than inflated headline numbers.

EDUCATION

M.S., Data Science (GPA 3.72)

Expected May 2027

George Washington University, Washington, DC

- *Relevant Coursework:* Data Warehousing, Data Mining, Algorithm Design for Data Science, Machine Learning I: Algorithm Analysis, Visualization of Complex Data, NLP for Data Science, Deep Learning and the Fundamentals of AI, Time Series Analysis & Modeling

B.A., Sport Management: Minor, Data Analytics (GPA 3.55)

Dec 2024

High Point University, High Point, NC

- *Relevant Coursework:* Applied Multivariate Statistics, Data Analysis & Predictive Analysis, Intermediate Statistical Analysis, Introduction to SAS Programming, Introduction to Statistical Analysis, Linear Algebra

TECHNICAL SKILLS

- *Languages:* Python, SQL, R, SAS, C++, JavaScript
- *ML & Data Science:* pandas, NumPy, SciPy, scikit-learn, LightGBM, XGBoost, PyTorch, SHAP, causalml, econml, Optuna, SMOTE
- *MLOps & Serving:* FastAPI, Pydantic, Docker, MLflow, GitHub Actions (CI), pytest, ruff, Render deployment, REST APIs
- *Visualization & Apps:* Streamlit, Plotly, Dash, Power BI, Tableau, Matplotlib, seaborn, ggplot2, Shiny/Leaflet
- *Modeling & Analytics:* Regression, Classification, Causal Inference / Uplift Modeling, Time-Series Forecasting & CV, Anomaly Detection, Sequential Hypothesis Testing (SPRT/CUSUM), Model Evaluation, Statistical Analysis

PROJECT EXPERIENCE

GridCast: Hourly Electricity Load Forecasting (MLOps)

Python, LightGBM, FastAPI, MLflow, Docker, GitHub Actions | 2026

- Built and served an hourly electricity-load forecaster that cut MAE 72% over a seasonal-naive baseline (2.67% vs 9.54% MAPE) on 18,169 held-out hours, using LightGBM with expanding-window time-series cross-validation and leakage-safe lag and rolling features.
- Shipped a FastAPI inference service that engineers features server-side from raw load history to eliminate train/serve skew, with MLflow experiment tracking, a Docker image, and a GitHub Actions CI pipeline running lint, unit tests, and end-to-end training and serving smoke-tests.

UpliftIQ: Uplift Modeling & Retention API

Python, causalml, econml, FastAPI, Docker, Render | 2026

- Built and deployed a causal uplift-modeling API that estimates the per-customer treatment effect (CATE) of a retention contact and routes budget to “persuadable” customers, served through FastAPI with Pydantic validation and a public dashboard hosted on Render.
- Benchmarked S-learner, T-learner, and Causal Forest by Qini coefficient on confounded observational data (propensity AUC ≈ 0.97), deployed the only learner to beat random targeting, and explicitly declined to report the naive treated-vs-control difference as a causal effect.

HouseEdge: Adversarial Game-Integrity & Advantage-Play Detection

Python, SciPy, scikit-learn, Dash, Render | 2026

- Built a two-sided game-integrity system that simulates fair and subtly-rigged casino games and detects both rigging and player advantage-play using sequential hypothesis testing (SPRT, CUSUM) and anomaly detection (Isolation Forest, Gradient Boosting), deployed as a live Dash surveillance console.
- Quantified a detectability frontier under adversarial adaptation — flagging a card counter ~100% of the time in a median ~83 hands at a 0.4% false-positive rate — and honestly exposed the detector’s blind spot by constructing a Wong back-counter that evades it entirely at a sustained +1% edge.

ClaimsGuard: Explainable Healthcare Fraud-Risk Prioritization

Python, scikit-learn, PyTorch, SHAP, Streamlit | 2026

- Built and deployed an explainable healthcare-claims anomaly detection system that tiers claims into a prioritized auditor worklist and pairs each flag with SHAP attributions in an interactive Streamlit dashboard.
- Used SHAP-based domain auditing to validate a feature-pruned Isolation Forest model, lifting top-tier fraud precision from 72% to 80% while enforcing leakage-safe evaluation and test-calibrated thresholds.

Parquet Capital: NBA Contract Valuation & Forecasting Engine

Python, scikit-learn, SHAP, PuLP, Streamlit | 2026

- Built an end-to-end ML pipeline to forecast NBA player performance trajectories with quantile-based uncertainty bands and classify contracts as overvalued, undervalued, or fairly valued.
- Applied expanding-window time-series cross-validation, calibrated prediction intervals to approximately 80% coverage, and stress-tested conclusions through cross-model validation, feature studies, and held-out dollar-impact analysis.

LungGuard: Explainable Lung Cancer Risk Prediction Dashboard

Python, scikit-learn, LightGBM, SHAP, Streamlit | 2026

- Built an end-to-end ML decision-support pipeline predicting lung cancer risk from 29 patient variables, pairing each prediction with SHAP explanations and a counterfactual “What-If” engine.
- Benchmarked Logistic Regression, Random Forest, and LightGBM with 5-fold stratified cross-validation, tuning the decision threshold to optimize F2 for screening recall and adding calibration, fairness auditing, and real-time risk stratification.

PROFESSIONAL EXPERIENCE

Analytics Intern, Division I Men’s Baseball Team

Jan – Apr 2024

High Point University – High Point, NC

- Analyzed Trackman pitch and batted-ball data for 30+ players to support player evaluation and performance review
- Evaluated team and opponent tendencies using Synergy and scouting workflows to support game preparation

Division I Men’s Basketball Team Manager

Aug 2022 – Nov 2023

High Point University – High Point, NC

- Conducted film analysis using HD Intelligence to identify high-efficiency shot zones

Film & Analytics Cohort

Aug – Sep 2023

Grow the Game, Hudl – High Point, NC

- Generated box scores and shot charts for 40+ basketball players and created five solo film segments using SportsCode (custom coding language)

Game Day Intern

May – Jul 2023

Winston-Salem Dash – Winston-Salem, NC

- Tracked live gameplay using DakStats for 20+ games